Employee Workforce Dashboard —Guide

Practical, step-by-step Excel tutorial based on the uploaded workbook

1. Prepare the workbook

- Open Excel desktop, enable editing only if you trust the workbook, and save a working copy as Excel Macro-Enabled Workbook (*.xlsm).
- Create three sheets named Employee_Data, Pivot and Dashboard. Keep the original source workbook as a separate backup.
- Place the 19 column headings in Employee_Data!A2:S2 and the 785 source records in rows 3:787.
- Select A2:S787, press Ctrl+T, tick My table has headers, and name the table tblEmployees under Table Design.
- Format Hire Date and Exit Date as dates, Annual Salary (INR) as INR currency, Age and Training Hours as numbers, and Attendance % as a numeric percentage scale.

2. Validate and prepare columns

- Use Data > Remove Duplicates on Employee ID only after checking whether duplicates represent legitimate records.
- Check missing Exit Date values against Employment Status; do not fill missing exit dates with invented dates.
- Create Year from Hire Date with =YEAR([@[Hire Date]]) and Month with =TEXT([@[Hire Date]],"mmm") if the reporting goal is hire-cohort analysis. These are reproducible example formulas; verify the original field semantics.
- For a reproducible tenure snapshot, place a documented as-of date in Settings or a fixed cell (e.g., Pivot!AF1). Use =YEARFRAC([@[Hire Date]],IF([@[Exit Date]]="",Pivot!$AF$1,MIN([@[Exit Date]],Pivot!$AF$1)),1) and format to 2 decimals. This is an example; the workbook's original tenure logic may differ.
- Check negative tenure, dates after the as-of date, missing salaries, and Attendance % outside 0–100.

3. Build the core PivotTables

- Select any cell in tblEmployees > Insert > PivotTable > Existing Worksheet > Pivot!D5.
- Drag Employment Status into Rows and Employee ID into Values. Set Values to Count, not Sum. The result should show Active and Attrition.
- Create a department count PivotTable with Department in Rows and Count of Employee ID in Values; place it near Pivot!G5.
- Create a department attrition PivotTable with Department in Rows, Employee ID in Values, and Employment Status filtered to Attrition; place it near Pivot!O5.
- Create a tenure PivotTable with Tenure in Values, change aggregation to Average, and place near Pivot!D17.
- Add separate PivotTables for location, job level, performance rating and hire month. Preserve the same source table for all PivotTables to simplify slicer connections.

4. Reproduce the KPI formulas

- In Pivot!A6:A10 enter Total Employee, Active Employee, Attrition, Attrition %, Avg Tenure.
- Pivot!B6: =GETPIVOTDATA("Employee ID",$D$5)

- Pivot!B7: =GETPIVOTDATA("Employee ID",$D$5,"Employment Status","Active")
- Pivot!B8: =GETPIVOTDATA("Employee ID",$D$5,"Employment Status","Attrition")
- Pivot!B9: =IFERROR(B8/B6,0), formatted as percentage. Original formula is =B8/B6.
- Pivot!B10: =GETPIVOTDATA("Tenure",$D$17), formatted to two decimals.
- Department attrition example: =IFERROR(GETPIVOTDATA("Employee ID",$O$5,"Department",A14)/GETPIVOTDATA("Employee ID",$G$5,"Department",A14),0). The workbook uses this pattern in Pivot!B14 and following rows.

5. Create charts and dashboard cards

- On Pivot, select a department-count PivotTable > PivotTable Analyze > PivotChart > Clustered Bar. Label chart Workforce by Department.
- Use a PivotChart for Attrition by Department based on the filtered attrition PivotTable; a bar chart supports department comparisons.
- Create a stacked column chart from monthly Active/Attrition counts, and a horizontal bar chart for Performance Rating counts.
- Create a location headcount bar chart and a job-level breakdown chart from their PivotTables. The original file has seven chart definitions; these instructions describe a reproducible layout, not an exact one-to-one mapping of all seven chart types.
- On Dashboard, insert rounded rectangles for KPI cards; select each shape and type =Pivot!B6 (and corresponding cells B7:B10) into the formula bar to link values.
- Move charts to Dashboard using Chart Design > Move Chart > Object in Dashboard; align with Shape Format > Align.
- Hide worksheet gridlines via View > Gridlines, apply consistent olive/amber palette, and add an explanatory subtitle and reporting-as-of date.

6. Create and connect slicers

- Click the status/headcount PivotTable > PivotTable Analyze > Insert Slicer. Select relevant fields such as Department, Location, Job Level, Employment Type and Employment Status. The workbook has five slicer caches, but exact original field-to-cache mapping should be checked in Excel.
- Select each slicer > Slicer tab > Report Connections (or PivotTable Connections).
- Tick every PivotTable that should respond to the slicer. PivotTables generally must share the same source/PivotCache for the connection to be available.
- Test by selecting one Department and verifying that KPI cards, charts and headcount totals all change consistently.
- Use Ctrl+click for multiple selections or the slicer Multi-Select control; use Clear Filter to reset.
- If a PivotTable is absent from Report Connections, recreate it from the same Excel Table/PivotCache rather than assuming it is connected.

7. VBA macros: example implementations

The uploaded file contains xl/vbaProject.bin, but the original VBA source could not be extracted and verified in this environment. The following VBA is newly supplied, reproducible example code, not a transcription of the workbook macros.

```
Option Explicit
Public Sub ToggleDashboardFullScreen()
    With Application
        .DisplayFullScreen = Not .DisplayFullScreen
        .DisplayFormulaBar = Not .DisplayFullScreen
```

```
    End With
    ActiveWindow.DisplayGridlines = False
End Sub

Public Sub RefreshWorkforceDashboard()
    On Error GoTo ErrHandler
    Application.ScreenUpdating = False
    ThisWorkbook.RefreshAll
    Application.CalculateUntilAsyncQueriesDone
    Application.CalculateFull
    Application.ScreenUpdating = True
    MsgBox "Dashboard refresh completed.", vbInformation
    Exit Sub
ErrHandler:
    Application.ScreenUpdating = True
    MsgBox "Refresh failed: " & Err.Description, vbExclamation
End Sub

Public Sub ExitDashboardFullScreen()
    Application.DisplayFullScreen = False
    Application.DisplayFormulaBar = True
End Sub
```

8. Attach VBA to dashboard buttons

- Save as .xlsm, then File > Options > Customize Ribbon > enable Developer.
- Press Alt+F11 > Insert > Module and paste the example code. Use Debug > Compile VBAProject.
- Return to Dashboard > Insert > Shapes; create buttons labeled Full Screen, Refresh and Exit Full Screen.
- Right-click each shape > Assign Macro > choose ToggleDashboardFullScreen, RefreshWorkforceDashboard or ExitDashboardFullScreen.
- Save, close, reopen, and enable macros only in a trusted copy. Test each button and verify refresh updates all relevant PivotTables.
- Full-screen changes application-level UI state; use Exit Full Screen before leaving the workbook.

9. Validation checklist

- Unfiltered Pivot totals must reconcile to 785 employees, 561 Active and 224 Attrition; average tenure should approximate 5.09 with original data logic.
- Compare slicer-filtered PivotTable counts with COUNTIFS checks against the source data.
- Confirm no #REF!, #DIV/0!, stale pivot caches or disconnected charts.
- Test Refresh after adding a synthetic row to tblEmployees, then remove the row.
- Inspect the workbook in Excel desktop, because Excel Online does not run VBA macros.
- Do not upload the original employee-level dataset to a public repository without anonymization and permission.

10. Publish to GitHub

- Create repository Excel-Employee-Workforce-Dashboard.
- Add README.md, project documentation PDF, this rebuild guide, and sanitized workbook.
- Add screenshots of the dashboard with filters reset; avoid displaying personal identifiers.
- Document Excel desktop version, macro trust requirements, source provenance and data limitations.